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Markov Models - Assessment
Graded Assignment • 15 min

Due Nov 16, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Why is it necessary to compute and use prior probabilities in building classifiers? 1 / 1 point

- To avoid using other forms of probability
- To prolong the classifier's operational lifespan
- To streamline the training process with less data
- To provide an accurate initial state for class prediction

✓ Nice work

Correct! Priors help determine the starting probability estimates for classes.

2. What is needed to form a Markov model from a sequence of data? 1 / 1 point

- Randomly selected initial state probabilities
- A sequence of outcomes without transition details
- Only outcome data
- A list of all possible states and rules for transitions

✓ Nice work

Correct! A Markov model requires a defined set of states and transition probabilities between them.

3. What makes classifying text an example of supervised learning? 1 / 1 point

- It requires human intervention during each step.
- The classifier operates without human oversight.
- The use of labeled data for training the classifier
- The model generates labels during training.

✓ Nice work

Correct! Supervised learning relies on labeled datasets to instruct algorithm development.

4. In a Markov model, why do transition probabilities sum to one in each row? 1 / 1 point

- Transition options from a state must cover all possibilities.
- The model is balanced by default.
- Rows are normalized for consistency, not probability.
- Each row should have one correct transition only.

✓ Nice work

Correct! Each state must transition to some other state, including itself, thus the probabilities sum to one.

5. What advantage do Markov models provide for modern NLP tasks? 1 / 1 point

- They simplify language into static word lists.
- They guarantee accurate comprehension of textual data.
- They eliminate the need for any learning in NLP.
- They provide a framework for state-transition modeling, aiding in sequence prediction.

✓ Nice work

Correct! Markov models create structured models for handling sequences predictively.